

1 Pre-lab design

1.1 Question D1

The following model describes the changes in electrical resistance as a function of strain:

$$\epsilon = \frac{dR}{RG'} \quad (1)$$

Where:

- ϵ represents strain factor
- R is the nominal resistance
- dR is the change in resistance
- G is the gauge factor.

Calculate the change in resistance dR if $R = 120\Omega$, $G = 2$ and $\epsilon = 0.01$:

$$\begin{aligned} dR &= \epsilon RG' \\ dR &= 0.01 \cdot 120\Omega \cdot 2 \\ dR &= 2.4\Omega \end{aligned} \quad (2)$$

1.2 Question D2

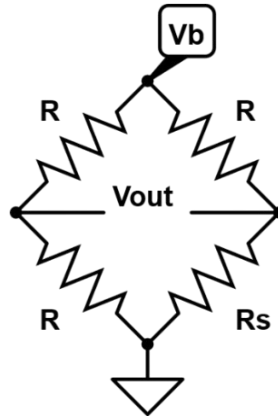


Figure 1: Wheatstone bridge.

Let the strain gauge resistance be represented by R_s :

$$R_s = R + dR \quad (3)$$

Let the other three resistors have the same nominal resistance R_s as the strain gauge. The expression for V_{out} becomes:

$$\begin{aligned}
V_{out} &= V_b \cdot \frac{R}{R+R} - V_b \cdot \frac{R_s}{R_s+R} \\
V_{out} &= V_b \left(\frac{1}{2} - \frac{R_s}{R_s+R} \right) \\
V_{out} &= V_b \left(\frac{1}{2} - \frac{R+dR}{2R+dR} \right)
\end{aligned} \tag{4}$$

1.3 Question D3

Consider $V_b = 5V$, $R = 120\Omega$, calculate V_{out} for a 1% strain:

$$\begin{aligned}
V_{out} &= V_b \cdot \frac{R}{R+R} - V_b \cdot \frac{R_s}{R_s+R} \\
V_{out} &= 5V \cdot \frac{120\Omega}{240\Omega} - 5V \cdot \frac{120\Omega + 2.4\Omega}{120\Omega + 120\Omega + 2.4\Omega} \\
V_{out} &= 5V \left(\frac{1}{2} - \frac{122.4\Omega}{242.4\Omega} \right) \\
V_{out} &= -0.02475V
\end{aligned} \tag{5}$$

Re-calculate now where each bridge resistor features $R = 114\Omega$:

$$\begin{aligned}
V_{out} &= V_b \cdot \frac{R}{R+R} - V_b \cdot \frac{R_s}{R_s+R} \\
V_{out} &= 5V \cdot \frac{114\Omega}{228\Omega} - 5V \cdot \frac{120\Omega + 2.4\Omega}{120\Omega + 114\Omega + 2.4\Omega} \\
V_{out} &= 5V \left(\frac{1}{2} - \frac{122.4\Omega}{236.4\Omega} \right) \\
V_{out} &= -0.08883V
\end{aligned} \tag{6}$$

1.4 Question D4

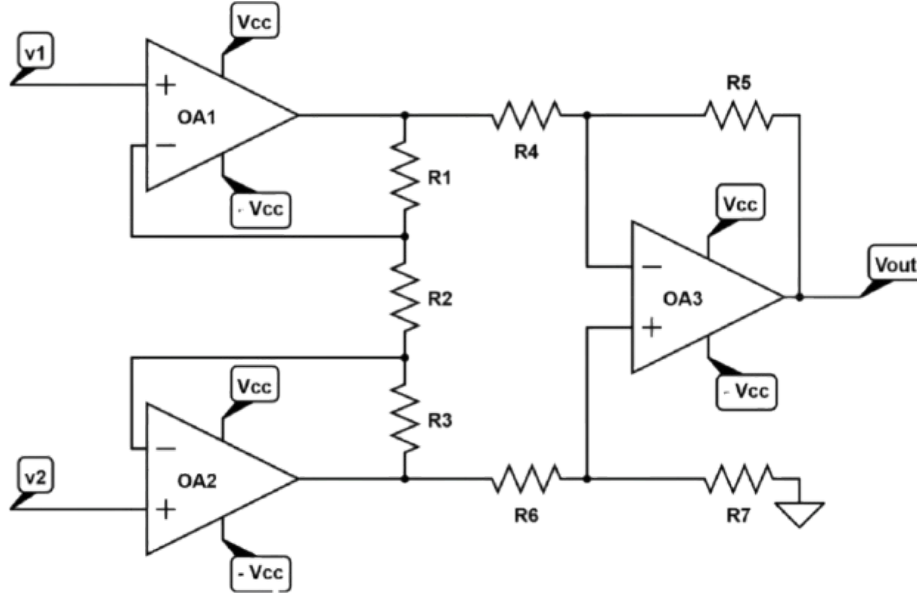


Figure 2: Instrumentation amplifier.

We wish to find an expression for $\frac{V_{out}}{V_{in}}$, Let us denote V_{OA1} , V_{OA2} as opamp 1 and 2's output voltages respectively:

$$I_{R_2} = \frac{V_1 - V_2}{R_2} \quad (7)$$

Notice that the current must be equal for $I_{R_1} = I_{R_2} = I_{R_3}$, therefore:

$$\begin{aligned} I_{R_1} &= \frac{V_{OA1} - V_1}{R_1} \\ V_{OA1} - V_1 &= I_{R_1} \cdot R_1 \\ V_{OA1} &= I_{R_1} \cdot R_1 + V_1 \\ V_{OA1} &= \frac{V_1 - V_2}{R_2} \cdot R_1 + V_1 \end{aligned} \quad (8)$$

This can similarly be repeated for V_{OA2} to get:

$$\begin{aligned} I_{R_3} &= \frac{V_2 - V_{OA2}}{R_3} \\ V_2 - V_{OA2} &= I_{R_3} \cdot R_3 \\ V_{OA2} &= V_2 - I_{R_3} \cdot R_3 \\ V_{OA2} &= V_2 - \frac{V_1 - V_2}{R_2} \cdot R_3 \end{aligned} \quad (9)$$

Let us denote the inputs of the final op-amp as V_+ , V_- , we then get:

$$\begin{aligned}
 V_{out} = & \left(V_2 - \frac{V_1 - V_2}{R_2} R_3 \right) \frac{R_7}{R_6 + R_7} \\
 & - \frac{R_5}{R_4} \left[\left(\frac{V_1 - V_2}{R_2} R_1 + V_1 \right) \right. \\
 & \left. - \left(V_2 - \frac{V_1 - V_2}{R_2} R_3 \right) \frac{R_7}{R_6 + R_7} \right]
 \end{aligned} \tag{10}$$

The above equation can be simplified into:

$$\begin{aligned}
 V_{out} = & \frac{V_2 [R_7(R_4 + R_5)(R_2 + R_3) + R_1 R_5(R_6 + R_7)]}{R_2 R_4 (R_6 + R_7)} \\
 & - \frac{V_1 [R_3 R_7(R_4 + R_5) + R_5(R_1 + R_2)(R_6 + R_7)]}{R_2 R_4 (R_6 + R_7)}
 \end{aligned} \tag{11}$$

1.5 Question D5